

STRUCTURED QUESTIONS (100 MARKS)

(INSTRUCTION: This paper has **SEVEN (7)** questions. The marks for each question is as indicated. Please write the answers for **ALL** questions in the provided answer booklet.)

QUESTION 1 [20 Marks]

- (a) Given the information provided below. The given values are in decimal format and must be assigned using directive **WORD**.

value_1=300, value_2=20, value_3=550,value_4=48

Write the complete assembly code to execute equation :

(value1+value2) * (value3-value4)

Initialize identifiers in data segment. Instruction MUL, registers EAX and EBX are allowed to be used. [6 Marks]

- (b) Given the contents of the following *DumpRegs* are as shown in **Figure 1**:

EAX=770C0810	EBX=7FFDC530	ECX=7080030A	EDX=00400000
ESI=00000000	EDI=00000000	EBP=0012FF94	ESP=0012FF8C
EIP=00401025	EFL=00000B82	CF=0	SF=1 ZF=0 OF=1

Figure 1.

From the answer in (a), determine the final content of register EAX, EBX and EDX after executing the instructions. Show your working clearly in multiplication line. [3 Marks]

- (c) Assuming that the system uses a word sized number and uses even parity. Show the calculation for the below equation in binary and determine the value for each of the flags. Write the flags value for ZF, SF, CF, OF, AF and PF. [4 Marks]

74CFh - 3A9Bh

- (d) Based on the Instruction Set Architecture (ISA) format as shown in **Figure 2**, answer the following questions:

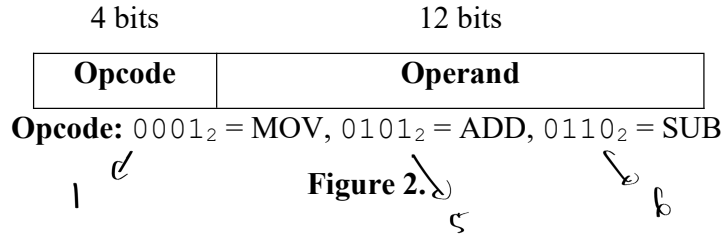


Table 1.

Memory Address	Memory Content	Instructions
56A	156A	MOV BX, [801]
56B	156B	MOV AX, BX
56C	656C	SUB AX, [802]
56D	556D	ADD AX, VAL2
...		...
801	9008	
802	089A	
803	B803	
...		...

Table 2.

Clock	IP/PC	MAR	MBR	IR	AX	Micro-Operations
t_0					9008	IP/PC = 56C

- i) Based on **Table 1**, analyze the content memory from address 56A to 56B. Determine the value of accumulator AX? [1 Mark]

$$AX = 9008$$

- ii) Trace and complete the data flow of instruction execution cycle in **Table 2** for instruction in address 56Ch only with its corresponding micro-operations. Write all the changes in the related CPU registers. [6 Marks]

QUESTION 2 [15 Marks]

- (a) Given the segments of a pipeline are listed in **Table 3** and the interface delay of register is 5 microseconds (μs). Answer the following questions and show your working clearly:

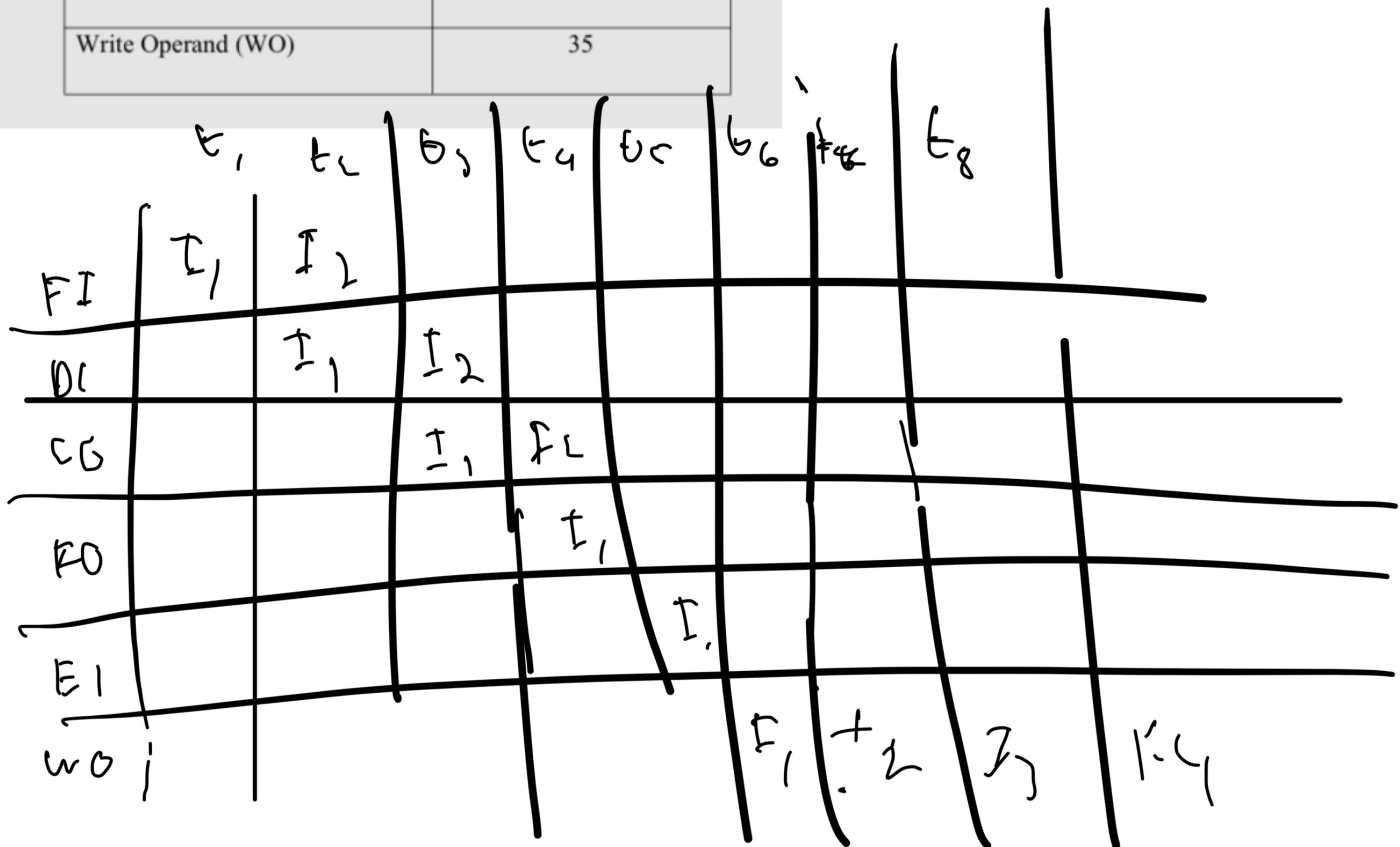
Table 3.

Segment	Execution Times (μs)
Fetch Instruction (FI)	50
Decode Instruction (DI)	45
Calculate Operand (CO)	30
Fetch Operand (FO)	40
Execute Instruction (EI)	45
Write Operand (WO)	35

- i) Draw a complete space time diagram on instruction pipeline for 4 instructions labeled as I1, I2, I3 and I4 . [2 Marks]
- ii) Calculate cycle time of non-pipeline and pipeline. [2 Marks]
- iii) Calculate execution time for 2 Kilo tasks for non-pipeline and pipeline. [3 Marks]
- iv) Calculate real Speedup, S. [2 Marks]
- v) Calculate maximum Speedup, S. [1 Mark]

Table 3.

Segment	Execution Times (μs)
Fetch Instruction (FI)	50
Decode Instruction (DI)	45
Calculate Operand (CO)	30
Fetch Operand (FO)	40
Execute Instruction (EI)	45
Write Operand (WO)	35



- (b) **Figure 3** illustrates instructions of an assembly program that will be implemented in pipeline using the following segments: Fetch Instruction (FI), Decode Instruction (DI), Execute Instruction (EI) and Write Operand (WO). Determine which instruction(s) have the following hazards. Justify your answer and rewrite the instructions using the proposed solution.

```
I1:          MOV      AX, 100
I2:          SUB      BX, AX
I3:          MOV      CX, VAL1
I4:          MOV      ESI, offset VAL2
I5:          INC      DX
I6:          JMP      TARGET
I7:          INC      CX
. . . . .
I12:  TARGET  DEC      AX
```

Figure 3.

- i) Data dependency and solve using delayed load. [2.5 Marks]
- ii) Branch difficulty and solve using rearranging instructions. [2.5 Marks]

QUESTION 3 [15 Marks]

- (a) Write the differences of Hardwired and Microprogrammed Control Units by completing **Table 4**, in terms of Structure, Operation, Flexibility and Performance. [4 Marks]

Table 4.

Characteristic	Hardwired Control Unit	Microprogrammed Control Unit
Structure	use the logic circuit	use control memory (ROM) contain micro instructions
Operation	generate control signal directly through hardware logic	generate control signal by executing micro instructions stored in control memory
Flexibility		high → microprogram can be modified or updated easily
Performance		slower due to fetching and executing microinstruction from control memory

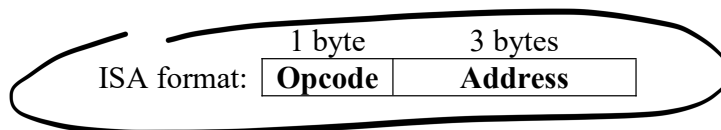
- (b) The ISA format for the instruction in **Figure 4** is defined in two fields which are Opcode and Address as shown in **Figure 5**. The first byte of the hex value in Byte Code represents the Opcode of the instruction. Using Direct Mapping method, determine the first address (in hex) of micro instruction routine for the following instruction in Control Memory. Given the Binary Mapping Bits in 6 bits as 0XXXX01. [2 Marks]

Instruction	Byte Code
ADD AX, VALUE	5941

Figure 4.

opcode

59 h = 0101 1001
replace with lower 4 bit



0 1001 01
= 25_h

Figure 5.

- (c) Given vertical micro-instruction format in 20 bits is given in **Figure 6** as follows:

No. of bits:	3	3	3	2	2	7
Field:	F1	F2	F3	CD	BR	AD

Figure 6.

Table 5 shows partial symbolic micro-program and **Table 6** shows micro-program implementation using ROM. Determine the binary values for F1, F2, F3, CD, BR, and AD for micro-operation READ and ADD at decimal address 1 and 2 in **Table 6**. Address micro-instruction **FETCH** routine in Control Memory start at decimal Address 64. Details of micro-instruction fields are described in the **Appendix**. [3 Marks]

Table 5.

Label	Micro-operation	CD	BR	Address
ADD:	ORG 0			
	NOP	I	CALL	INDRCT
	READ	U	JMP	NEXT
	ADD	U	JMP	FETCH

→ next 12

Table 6.

Micro-routine	Address		Binary Micro-instruction					
	Decimal	Binary	F1	F2	F3	CD	BR	AD
ADD	0	0000000						
	1	0000001	000	000	000	00	00	0000 001
	2	0000010	000	000	000	00	00	000 0000
	3	0000011						

- (d) A Control Unit uses a micro-instruction format consisting of 40 bits. The Micro-operation field uses 20 bits, the Next-address field uses 12 bits, and the Condition Select field uses 5 bits. Show your working clearly.

- i) Calculate how many Control Memory words that can be addressed. [2 Marks]

$$2^{12} = 4096$$

- ii) Determine how many unique conditions can be tested with the given Condition Select field. [2 Marks]

$$2^5 = 32$$

32 unique condition

- iii) Calculate how many bits remain unused or available for future use in the micro-instruction. [2 Marks]

$$40 - 20 - 12 - 5 = 3 \text{ bits}$$

QUESTION 4 [10 Marks]

- (a) Given the memory address size, answer the questions and show your working clearly:
- i) 24 bits memory address, each address can hold 24 bits. Identify the memory capacity? [3 Marks]
 - ii) 32 Mbytes memory is divided into 64 Kbits blocks. How many blocks available? [3 Marks]
- (b) Given a memory address is 34Ah (10 bits) and there are 4 memory banks. Determine the memory bank address and the address of the word in the bank using LOI. [4 Marks]

QUESTION 5 [15 Marks]

- (a) Suppose a main memory of 2^{14} bytes is map to a 4-way Set Associative for 32 blocks, where each block contains 16 words. Show your working clearly.
- i) How many bits used for the **tag** and **set** for the main memory address? [4 Marks]
 - ii) Consider a word at the memory address 1BA6h need to be read. Which **set** and **tag** in the cache memory will be accessed? [4 Marks]
- (b) Assume that 43% of the instructions in a program are data accesses. The cache hit ratio is 76% and the hit time is 3 cycles, but the miss penalty is 15 cycles. Calculate the Average Memory Access Time (AMAT). [3 Marks]
- (c) Suppose a single cache fronting a main memory, which has 250 nanosecond access time, and the cache memory has access time 15 nanoseconds. Calculate the Effective Access Time (EAT) if the hit miss rate is 0.33. [4 Marks]

QUESTION 6 [10 Marks]

- (a) Identify the most accurate term for (i) to (v) in **Table 7**. Select the term from the given options in **Figure 7**. [5 Marks]

Table 7.

Description	Term
Polling wastes the processor's time because the processor is faster than the I/O devices.	(i)
It was developed to improve data reliability through the implementation of Hamming code for error detection and correction.	(ii)
CPU is alerted by an interrupt signal when an I/O device is ready, instead of constantly checking it.	(iii)
Memory disks that consist of platters, spindle, read/write heads, actuator arm, and controller.	(iv)
Use a separate processor to manage data transfer between memory and I/O devices, freeing the CPU for other tasks.	(v)

Channel I/O	Interrupt Driven I/O	Programmed I/O
Direct Memory Address (DMA)	Interrupt flag	Parity flag
Hard Disk Drives	RAID Level 6	Floppy Disks
		RAID Level 2

Figure 7.

- (b) **Table 8** shows the step-by-step procedure for writing data to the Disk Drives using DMA in a random sequence. Rearrange the subsequent steps in the correct order to guarantee the successful completion of the writing data, by writing the Step number (1 to 5) in (i) to (v). [2.5 Marks]

Table 8.

Step	Step number
DMA transfers data and subsequently deactivates the Request.	(i)
Once Ready is asserted, no additional devices may access the bus.	(ii)
The controller transfers data to the disk buffer and disables Ready.	(iii)
The controller identifies its address, activates disk control, and indicates readiness if the disk is prepared.	(iv)
The DMA transmits the disk controller's address and activates the Request and Write signals.	(v)

- (c) Determine the specific bit required for data regeneration labeled as (a) to (e) in a server employing RAID Level as shown in **Figure 8**. [2.5 Marks]

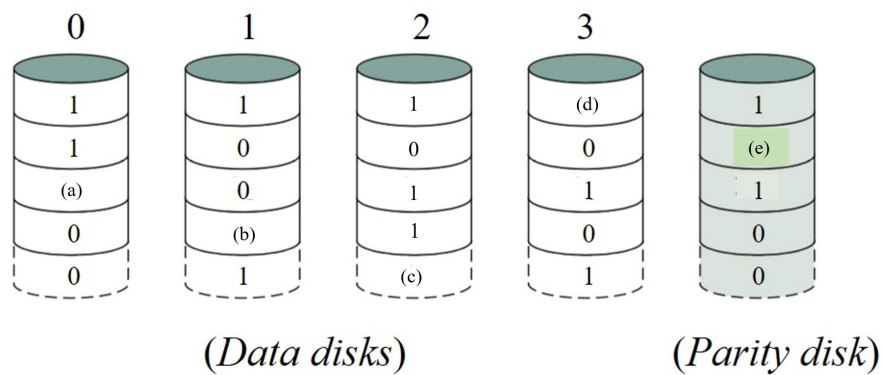


Figure 8.

QUESTION 7 [15 Marks]

- (a) **Table 9** shows the details of a Program A which consists of 155 instructions, executed on CPU X. Based on this details, answer the following questions and show your working clearly:

Table 9.

Instruction Class	Instructions Count	Clock Cycles
Load/Store	50	2
Arithmetic	40	4
Input/Output	30	6
Branch	20	5
Boolean	15	3

- i) Calculate the CPI for each instruction class given below. [2.5 Marks]
- ii) Calculate the Execution Time for each instruction class, given a clock cycle time is 0.2 microseconds. [2.5 Marks]
- iii) Calculate the CPU X's Execution Time. [1 Mark]
- iv) Calculate CPU X's Clock Rate. [2 Marks]
- v) Calculate the new CPU X's Execution Time if its Clock Rate is doubled and determine whether the CPU X become faster or slower. [2 Marks]

- (b) **Table 10** shows the Execution frequency and time (in picoseconds) for 3 different programs run by CPU P, Q and R.

Table 10.

Program	Execution frequency (%)	Execution time (<i>ps</i>)		
		CPU P	CPU Q	CPU R
A	35	250	100	400
B	40	300	400	350
C	25	50	200	150

- i) Calculate the Weighted Average for CPU P, Q and R. [3 Marks]
- ii) Choose the fastest CPU and how many percentage it is faster than the slowest CPU? [2 Marks]

----- End of Questions -----

Appendix

Micro-instruction Field Description

CD	Condition	Symbol	Comments
00	Always = 1	U	Unconditional branch
01	DR(15)	I	Indirect address bit
10	AC(15)	S	Sign bit of AC
11	AC = 0	Z	Zero value in AC

CD

BR	Symbol	Function
00	JMP	CAR $\leftarrow$ AD if condition = 1 CAR $\leftarrow$ CAR + 1 if condition = 0
01	CALL	CAR $\leftarrow$ AD, SBR $\leftarrow$ CAR + 1 if condition = 1 CAR $\leftarrow$ CAR + 1 if condition = 0
10	RET	CAR $\leftarrow$ SBR (Return from subroutine)
11	MAP	CAR(2-5) $\leftarrow$ DR(11-14), CAR(0,1,6) $\leftarrow$ 0

BR

F1	Microoperation	Symbol
000	None	NOP
001	AC $\leftarrow$ AC + DR	ADD
010	AC $\leftarrow$ 0	CLRAC
011	AC $\leftarrow$ AC + 1	INCAC
100	AC $\leftarrow$ DR	DRTAC
101	AR $\leftarrow$ DR(0-10)	DRTAR
110	AR $\leftarrow$ PC	PCTAR
111	M[AR] $\leftarrow$ DR	WRITE

F1

F2	Microoperation	Symbol
000	None	NOP
001	AC $\leftarrow$ AC - DR	SUB
010	AC $\leftarrow$ AC $\vee$ DR	OR
011	AC $\leftarrow$ AC $\wedge$ DR	AND
100	DR $\leftarrow$ M[AR]	READ
101	DR $\leftarrow$ AC	ACTDR
110	DR $\leftarrow$ DR + 1	INCDR
111	DR(0-10) $\leftarrow$ PC	PCTDR

F2

F3	Microoperation	Symbol
000	None	NOP
001	AC $\leftarrow$ AC $\oplus$ DR	XOR
010	AC $\leftarrow$ AC'	COM
011	AC $\leftarrow$ shl AC	SHL
100	AC $\leftarrow$ shr AC	SHR
101	PC $\leftarrow$ PC + 1	INCPC
110	PC $\leftarrow$ AR	ARTPC
111	Reserved	

F3